LESSON 01 / 03

Status of this sheet

Part I derived total, reflexive, symmetric and reflexive+symmetric counts with direct pair analysis. This sheet completes the catalogue (antisymmetric, equivalence/Bell, surjections by inclusion–exclusion) and shows the small-case listings an IOQM/RMO solution must carry. All counts are **models built from the Part I supply list** (product rule, complements, inclusion–exclusion on sets) — the technique is exam-derivable even where the topic is listed as an extension.

The catalogue for $|A| = n$

KEY RESULT

On an n-element set: relations $= 2^{n^2}$; reflexive $= 2^{n^2-n}$; symmetric $= 2^{n(n+1)/2}$; antisymmetric $= 2^n \cdot 3^{n(n-1)/2}$; reflexive+symmetric $= 2^{n(n-1)/2}$; reflexive+transitive = number of "preorder with fixed diagonal" (no simple closed form — casework); equivalence relations $= B_n$ (Bell): $1, 2, 5, 15, 52$ for $n = 1..5$. Functions $A \to B$ with $|B| = m$: m^n; injective: $\frac{m!}{(m-n)!}$ if $n \le m$ else 0; surjective: $n!\, S(n,m)$ with S the Stirling number $= \sum_{i=0}^{m} (-1)^i \binom{m}{i} (m-i)^n$.

SHORT DERIVATION

Surjections by inclusion-exclusion

$f : A \to B$ surjective iff no element of B is missed. Let U = all m^n functions and E_b = functions missing the value b. We want $|U \setminus \bigcup_b E_b|$. Now $|E_b| = (m-1)^n$ (values chosen from the rest), $|E_b \cap E_c| = (m-2)^n$, and so on, so inclusion–exclusion gives $\sum_{i=0}^{m} (-1)^i \binom{m}{i} (m-i)^n$. Small case $n = 3, m = 2$: $2^3 - 2 \cdot 1^3 = 6$ ✓ matches the 6 onto functions $\{1,2,3\} \to \{a,b\}$ found by cases in Part I. Sanity: $m > n$ gives 0 (e.g. $n = 2, m = 3$: $9 - 3 \cdot 4 + 3 \cdot 1 = 0$).

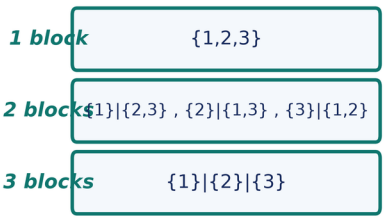
WORKED EXAMPLE

Worked. Onto functions $\{1,2,3,4\} \to \{a,b,c\}$: $3^4 - \binom{3}{1}2^4 + \binom{3}{2}1^4 - \binom{3}{3}0^4 = 81 - 48 + 3 =$. Cross-check with Stirling: $S(4,3) = 6$ (partitions of 4 elements into 3 blocks: one block of 2 and two singletons, $\binom{4}{2} = 6$), then order the blocks onto a, b, c: $6 \cdot 3! = 36$ ✓. Two independent methods agreeing is the standard olympiad "check" line.

Bell numbers by listings (do this in solutions)

For $A = \{1,2,3,4\}$, $B_4 = 15$ partitions: 1 block (1 way), 2 blocks: choose the block containing 1 ($S(4,2) = 7$), 3 blocks ($S(4,3) = 6$), 4 singletons (1). Total $1 + 7 + 6 + 1 = 15$ ✓.

General recurrence (extension): $B_{n+1} = \sum_{k=0}^{n} \binom{n}{k} B_k$ — element $n+1$ joins a block after k others are already placed... more cleanly: $B_{n+1} = \sum_{k=0}^{n} \binom{n}{k} B_{n-k}$: choose which elements share the block of $n+1$.



The 5 equivalence relations on {1,2,3} as partitions (Bell B_3 = 5)

COMMON TRAP

- **Labelled vs unlabelled.** $\{\{1,2\},\{3,4\}\}$ and $\{\{1,3\},\{2,4\}\}$ are different partitions of a labelled set; counting unlabeled block *shapes* is a different (smaller) problem. State which one you count.
- Antisymmetric count $2^n 3^{n(n-1)/2}$: each off-diagonal unordered pair $\{a,b\}$ allows exactly 3 states — neither direction, only $a \to b$, only $b \to a$ — never 4 (both directions are banned).
- Inclusion–exclusion signs alternate. Boundary convention: a function on an empty domain is unique (the empty function), so $m^0 = 1$ for every m, including $0^0 = 1$ — state the convention whenever $n = 0$ appears.
- Surjections need $n \geq m$; injections need $n \leq m$. A formula that spits out a nonzero answer violating these is being used wrong.

ASSUMPTIONS & CONVENTIONS

Counts assume **finite** sets and labelled elements throughout. $S(n,m)$ here means Stirling numbers of the second kind (onto functions divided by $m!$); the notation is Part II vocabulary, but every *numerical* claim above is checkable by listing for $n, m \leq 4$ — an olympiad write-up should always include one such listing.

Status of this sheet

The equations below are **additional theory** (classic RMO/INMO and AIME-adjacent material). Nothing here contradicts the syllabus: every *model* shown is solvable with the Part I supply list plus one extra idea, stated openly. Do not claim such an equation "can never appear in JEE" — exam papers have historically leaked olympiad-flavoured models; what is *not* promised is any theorem beyond the statements given here.

The Cauchy equation $f(x+y) = f(x) + f(y)$ on $\mathbb{R}$

SHORT DERIVATION

Rational linearity

$f(0) = f(0+0) = f(0) + f(0) \Rightarrow f(0) = 0$.

Induction: $f(nx) = nf(x)$ for $n \in \mathbb{N}$; then $0 = f(0) = f(x + (-x)) = f(x) + f(-x) \Rightarrow f(-x)$, so $f(nx) = nf(x)$ for all integers n. For p/q with $q \geq 1$:

$f(p) = f\left(q \cdot \frac{p}{q}\right) = qf\left(\frac{p}{q}\right) \Rightarrow f\left(\frac{p}{q}\right) = \frac{p}{q} f(1)$. Hence $f(r) = cr$ for every rational r, where $c = f(1)$.

KEY RESULT

Regular upgrade. If f is additionally monotone on any interval, or continuous at a point, or bounded above/below on any interval, then $f(x) = cx$ for all real x (rational-linearity plus density of $\mathbb{Q}$ and the squeeze). Without *some* regularity, non-linear solutions exist (via a Hamel basis; they are non-measurable and cannot be written down explicitly) — so: **never write $f(x) = cx$ as "the" solution unless the problem grants a regularity condition.** Many exam models quietly assume continuity; state it.

LESSON 02 / 03

WORKED EXAMPLE

Worked models.

1. $f(x+y)=f(x)+f(y)$ for all $x,y\in\mathbb{Q}$, $f(1)=2$; then $f(7/3)=\frac{7}{3}\cdot 2=\frac{14}{3}$ — no regularity needed, because the domain is $\mathbb{Q}$ itself (rational linearity already complete).
2. $f:\mathbb{R}\to\mathbb{R}$ continuous, $f(x+y)=f(x)+f(y)$, $f(2)=6$; $c=f(1)=3$, so $f(x)=3x$ for every real x. **Check:** $f(x+y)=3(x+y)=3x+3y$ ✓.
3. $f:\mathbb{R}\to\mathbb{R}$ with $f(x+y)=f(x)+f(y)$ and $f(x)\geq 0$ for all $x>0$ (bounded below on $(0,\infty)$); the regularity condition is what upgrades $f(r)=cr$ to $f(x)=cx$; then $f(x)\geq 0$ for $x>0$ forces $c\geq 0$.

The multiplicative cousins

KEY RESULT

$f(x+y)=f(x)f(y)$: either $f\equiv 0$, or $f(0)=1$ and f never vanishes; with continuity $f(x)=a^x$, $a=f(1)>0$. $f(xy)=f(x)+f(y)$ on $(0,\infty)$: with continuity $f(x)=c\ln x$ (set $g(t)=f(e^t)$ and Cauchy reduces). $f(xy)=f(x)f(y)$ on $(0,\infty)$: $f(x)=x^c$ or $f\equiv 0,1$ specials. **Pexider:** $f(x+y)=g(x)+h(y)$ for all x,y implies all three are affine with $f=g+h+\text{const}$ relationships — plug $y=0$ to express g,h through one function and fall back to Cauchy. **Jensen:** $f\left(\frac{x+y}{2}\right)=\frac{f(x)+f(y)}{2}$ with continuity $\Rightarrow f(x)=cx+d$.

SHORT DERIVATION

$f(x+y)=f(x)f(y)$ quickly

Put $y=0$: $f(x)=f(x)f(0)$, so either $f\equiv 0$ or $f(0)=1$. If $f(x_0)=0$ for some x_0, then $f(x)=f((x-x_0)+x_0)=f(x-x_0)f(x_0)=0$ for all x: $f\equiv 0$. So a non-zero solution never vanishes; if it takes both signs it still can't cross 0 (continuous image of connected set), hence a positive solution has $f(x)=e^{g(x)}$ with g additive, and continuity finishes $g(x)=x\ln a$.

Involutions $f(f(x))=x$

$f\circ f=\mathrm{id}$ means f is its own inverse: f is bijective and $f^{-1}=f$. Solutions of $f(f(x))=x$ split into **fixed points** ($f(x)=x$) and **2-cycles** ($f(x)=y\neq x$, $f(y)=x$).

SHORT DERIVATION

Increasing involution fixes everything

Suppose f is strictly increasing and $f(f(x))=x$. If $f(x_0)>x_0$ then applying the increasing f gives $f(f(x_0))>f(x_0)>x_0$, i.e. $x_0>f(x_0)$ — contradiction. Similarly $f(x_0)<x_0$ contradicts. So $f(x)=x$ for all x: every increasing involution is the identity on its domain. A decreasing involution (e.g. $f(x)=a-x$, $f(x)=1/x$ on $\mathbb{R}^+$) instead swaps pairs around fixed points.

WORKED EXAMPLE

Worked. Find all continuous $f:\mathbb{R}\to\mathbb{R}$ with $f(f(x))=x$ and f decreasing. Any x with $f(x)=x$ is a fixed point; $g(x)=f(x)-x$ is decreasing (f dec, $-x$ dec... sum of two decreasing is decreasing) hence g has at most one zero — at most one fixed point. Example $f(x)=a-x$ has exactly one ($x=a/2$). The general decreasing-continuous involution picture: one fixed point (if g changes sign) and the rest of the line paired off in 2-cycles symmetric about it. **Check:** $f(x)=1/x$ on $(0,\infty)$: decreasing, $f(f(x))=x$ ✓, fixed point $x=1$ ✓.

COMMON TRAP

- Cauchy with domain $\mathbb{Q}$: rational linearity is the complete answer. Cauchy with domain $\mathbb{R}$: rational linearity + "regularity" is required; state the condition.
- $f(x)=x^2$ is even, $f(x)=x^3$ odd, but $f(x)=x^2+x^3$ is neither — decomposition (Part I) is the safe tool.
- In $f(f(x))=x$ problems, plugging $x=f(t)$ is the standard trick: $f(f(f(t)))=f(t)$, i.e. $f^3=f$; combined with $f^2=\mathrm{id}$ this just re-proves bijectivity — write the injection/surjection argument once, cleanly.
- Periodic + additive: $f(x+T)=f(x)$ with additivity forces $f(T)=f(0)=0$, and with continuity that collapses $f\equiv 0$ (since $f(r)=cr$ and $cT=0$). Without continuity a wild additive T-periodic solution can still exist on a Hamel basis — so again: state the regularity condition before claiming collapse. The point stands: ***imposing two strong structures usually collapses the family to trivial members, but "usually" needs a hypothesis.***

LESSON 03 / 03

Status of this sheet

The greatest-integer function is ***in the official syllabus*** (listed under special functions); Hermite's identity and the fixed-point decomposition are extra theory. Every worked model below is solved with syllabus tools only — so treat the identities as accelerators, and the models as fair game.

Hermite's identity

SHORT DERIVATION

Sum of n shifted floors

For real x and positive integer n:

$$\sum_{k=0}^{n-1}\left\lfloor x+\tfrac{k}{n}\right\rfloor=\lfloor nx\rfloor$$

Write $x=m+\theta$ with $m=\lfloor x\rfloor$, $\theta\in[0,1)$. Then $\lfloor x+\frac{k}{n}\rfloor=m+\lfloor\theta+\frac{k}{n}\rfloor$, so the left side is $nm+\sum_k\lfloor\theta+\frac{k}{n}\rfloor$. Now $\lfloor\theta+\frac{k}{n}\rfloor=1$ exactly when $\frac{k}{n}\ge 1-\theta$, i.e. $k\ge n(1-\theta)$; the number of $k\in\{0,\dots,n-1\}$ with $k\ge n(1-\theta)$ is $\lfloor n\theta\rfloor$ (boundary $n(1-\theta)$ an integer counts k from that value up). So the sum $=nm+\lfloor n\theta\rfloor=\lfloor nm+n\theta\rfloor=\lfloor nx\rfloor$ ✓.

WORKED EXAMPLE

Worked checks. $n=3$, $x=1.7$: $\lfloor 1.7\rfloor+\lfloor 1.7+0.33\rfloor+\lfloor 1.7+0.67\rfloor=1+2+$ ✓. $x=2$ exactly: $2+2+2=6=\lfloor 6\rfloor$ ✓.
Check boundary: when $n\theta$ is an integer the threshold $k\ge n(1-\theta)$ is attained (included) — $\lfloor\cdot\rfloor$ inequality is $\ge$ on both sides of the threshold count.

Floor identities and equations

READ IN ORDER: top to bottom in the left column, then top to bottom in the right column; then next sheet.

Floor identities and equations

KEY RESULT

$[x] \le x < [x]+1$;
$[x]+[y] \le [x+y] \le [x]+[y]+1$;
$[x]+[-x] = 0$ *if* $x \in \mathbb{Z}$ *else* -1;
$\{x\}+\{-x\} = 0$ *or* 1 *with the same split;*
number of integers in $(a, b]$ *is* $[b]-[a]$;
$[x] = c \Leftrightarrow x \in [c, c+1)$; $[x] < c \Leftrightarrow x < c$ *(for integer* c*);* $[x] \ge c \Leftrightarrow x \ge c$ *(integer* c*).*

STANDARD APPLICATION

Model (fully syllabus-level). *Solve* $\lfloor x^2 \rfloor = 2$ *over the reals. Sandwich:* $2 \le x^2 < 3$*. Left half:* $x^2 \ge 2 \Rightarrow |x| \ge \sqrt{2}$*; right half:* $x^2 < 3 \Rightarrow |x| < \sqrt{3}$*. Solution set* $= \{x : \sqrt{2} \le |x| < \sqrt{3}\} = [-\sqrt{3}, -\sqrt{2}] \cup [\sqrt{2}$ *—* ***closed*** *at* $\pm\sqrt{2}$ *(*$x^2 = 2$ *gives* $\lfloor 2 \rfloor = 2$ ✓*) and* ***open*** *at* $\pm\sqrt{3}$ *(*$x^2 = 3$ *would give* $\lfloor 3 \rfloor = 3$ ✗*). The answer is a set of reals, not a number. Integer version:* $x \in \mathbb{Z}$ *needs* $x^2 \in [2, 3)$*, but squares of integers are* $0, 1, 4, \ldots$ *—* ***no solutions.*** ***Check:*** *the two phrasings ("all real* x*" vs "all integer* x*") have completely different answers; the quantifier is part of the question.*

Fixed points and involutions (counting flavour)

$f(x) = x$ *is a* ***fixed point****;* $f(f(x)) = x$ *is an* ***involution equation*** *whose solution set decomposes into fixed points of* f *plus 2-cycles* $\{x, f(x)\}$ *with* $f(x) \ne x$*.*

KEY RESULT

If f *is strictly increasing and* $f(f(x)) = x$ *on an interval, then* $f(x) = x$ *everywhere on it (Part II sheet 02 proof). If* f *is strictly decreasing and continuous,* $g(x) = f(x) - x$ *is strictly decreasing, so* f *has* ***at most one*** *fixed point; each non-fixed* x *pairs with* $f(x)$ *in a 2-cycle. Counting solutions of* $f(f(x)) = x$ *for a concrete monotone* f *therefore reduces to: find fixed points (solve* $f(x) = x$*), then account for the symmetric 2-cycles.*

WORKED EXAMPLE

Worked. $f(x) = 3 - x$ *on* $\mathbb{R}$*: decreasing,* $f(f(x)) = 3 - (3 - x) = x$ ✓ *for* ***every*** x *— the entire line solves the involution equation, yet only* $x = 3/2$ *is a fixed point. Moral: "solutions of* $f(f(x)) = x$*" and "fixed points of* f*" are different sets; the question wording decides which is asked. With* $f(x) = \frac{1}{x}$ *on* $(0, \infty)$*:* $f(f(x)) = x$ *everywhere; fixed point* $x = 1$ *only.* ***Check:*** $f(2) = 1/2$*,* $f(1/2) = 2$ *— a 2-cycle, as predicted by the decreasing-case analysis.*

COMMON TRAP

- $\lfloor x \rfloor$ of a negative number: $\lfloor -1.2 \rfloor = -2$, not -1 (greatest integer **less or equal**). Truncation toward zero is a different function.
- In floor equations, the answer set's endpoints are decided by strict vs non-strict inequalities — write the sandwich first, then convert to interval notation, then test both endpoints.
- $\lfloor x^2 \rfloor = 2$ vs $\lfloor x \rfloor^2 = 2$: the second means $\lfloor x \rfloor = \pm\sqrt{2}$, impossible for integer $\lfloor x \rfloor$ — no solutions. Composition order matters even inside floor equations.
- Never square an inequality $\sqrt{2} \le |x| < \sqrt{3}$ without tracking that $|x|$ is already non-negative — squaring is safe on $[0, \infty)$ but flips nothing here; for $x < 0$ reflect by $|x|$.

ASSUMPTIONS & CONVENTIONS

All floor statements are for real x; n is a positive integer in Hermite; fixed-point claims assume the stated monotonicity/continuity on an interval. 2-cycles are unordered pairs $\{x, y\}$ with $y = f(x) \neq x$; when "count solutions" is asked on a finite set, count elements x, not cycles.